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SEMI E90-1224
SPECIFICATION FOR SUBSTRATE TRACKING

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1 Purpose

1.1 The purpose of this Specification is to provide the standard services of equipment to track substrates (manufactured product) in manufacturing equipment. This Specification defines the concepts and behaviors for the information management of substrates, as well as the messages/services.

2 Scope

2.1 Essentially, information about substrates must be managed by the factory system, while the equipment is required to provide the services for the substrate information management. This Specification addresses the requirement for the equipment services to manage information of substrates that reside in the equipment.

2.2 The scope of this Specification is to define the information services of equipment that can be requested by the user. To clarify required services, the concepts and behaviors of the substrate, the substrate location, the batch and the batch location are defined.

2.3 This Specification is applicable to any manufacturing equipment that handles substrates. To implement these services, the equipment and factory system must be integrated by means of a communication link.

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3 Limitations

3.1 This Specification assumes that the substrate(s) to be managed has been resident in the equipment and has been given a substrate identifier.

3.2 The service for a substrate is valid while the substrate is registered in the equipment. That is, the service is available from the moment that the substrate is registered into the equipment and, to the moment that the substrate is removed from the equipment.

3.3 This Specification is only for information services. It does not address any mechanism to read or write information from/to the substrate.

4 Referenced Standards and Documents

4.1 *SEMI Standards and Safety Guidelines*

SEMI E30 — Specification for the Generic Model for Communications and Control of Manufacturing Equipment (GEM)

SEMI E39 — Specification for Object Services: Concepts, Behavior, and Services

SEMI E172 — Specification for SECS Equipment Data Dictionary (SEDD)

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5 Terminology

5.1 *Definitions*

5.1.1 *batch* — a group of substrates to be processed in a process resource simultaneously.

5.1.2 *batch container* — a supporting structure that is used to hold substrates for processing, and it may visit multiple locations in equipment with substrates in it. Whether a batch container is used or not depends on the type of equipment. Typical example for a batch container is a 'boat' used in furnace equipment.

5.1.3 *batch location* — locations in the equipment where substrates visit as a group of substrates for storage or processing.

5.1.4 *carrier ID* — the name to identify a specific substrate carrier.

5.1.5 *carrier slot* — physical location capable of holding a substrate within cassette type carrier.

5.1.6 *carrier slot map* — the registry of substrates to the substrate carrier slots.

5.1.7 *carrier substrate location* — a substrate location within a substrate carrier capable of holding a substrate.

5.1.8 *default substrate ID (default ID)* — the substrate ID assigned to the substrate when no substrate ID information is given by the user but the carrier ID for the source carrier is known. The default ID is the combined text of the source carrier ID and the slot number.

5.1.9 *equipment substrate location* — a substrate location on an equipment resource.

5.1.10 *location ID* — the name of a material location.

5.1.11 *lot* — a group of one or more substrates of the same type. A lot must be organized by the user. The group may be referred to for tracking of substrates in the factory.

5.1.12 *material location* — an identifiable place within the equipment or carrier where material can be held.

5.1.13 *register* — an operation that adds the substrate object to the equipment's database. This operation is performed automatically when the equipment receives both a carrier and information from the host about the contents of the carrier. The operation is also performed automatically when the equipment detects the substrate ID.

5.1.14 *remove* — the operation that removes a substrate from the equipment.

5.1.15 *service* — represents a function offered to a user by a provider. A service consists of a sequence of service primitives, each described by a list of parameters.

5.1.16 *substrate* — the basic unit of material on which work is performed to create a product. Examples include wafers, die, plates used for masks, flat panels, circuit boards, and leadframes.

5.1.17 *substrate carrier* — a carrier to hold substrates to be transferred to/from the equipment. A substrate carrier has one or more position to hold substrates (carrier substrate location).

5.1.18 *substrate history* — ordered set of information about the locations visited by the substrate.

5.1.19 *substrate ID* — identifier of a substrate.

5.1.20 *substrate location* — a material location which is capable of holding a substrate. For example, but not limited to, process modules, transfer subsystems, wafer chucks, robot end effecter, and carrier slots.

5.1.21 *substrate type* — represents the type of the substrate, such as wafers, CDs, flat panels, or masks.

5.2 *Data Type*

5.2.1 *Boolean* — may take on one of two possible values, equating to TRUE or FALSE.

5.2.2 *enumerated* — may take on one of a limited set of possible values. These values may be given logical names, but they may be represented by any single-item data type except floating point.

5.2.3 *floating point* — may take on any single numeric value, positive or negative. Messaging protocol may impose a limit on the range of possible values.

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5.2.4 *form* — type of data: positive integer, unsigned integer, integer, floating point, enumerative, Boolean, text, formatted text, structure, list, and ordered list.

5.2.5 *formatted text* — text with an imposed format. This could be by position, by use of special characters, or both.

5.2.6 *integer* — may take on the value of any negative or unsigned integer. Messaging protocol may impose a limit on the range of possible values.

5.2.7 *list* — a set of one or more items that are all of the same form (one of the above forms).

5.2.8 *ordered list* — a list for which the order in which items appear is significant.

5.2.9 *positive integer* — may take the value of any positive whole number. Messaging protocol may impose a limit on the range of possible values.

5.2.10 *structure* — a complex set of information consisting of specific sets of items of possibly mixed data types, in a specified arrangement.

5.2.11 *text* — a text string. The message protocol restricts its length or ASCII representation. Messaging protocol may impose restrictions, such as length or ASCII representation.

5.2.12 *unsigned integer* — may take the value of any positive integer or zero. Messaging protocol may impose a limit on the range of possible values.

6 Convention

6.1 *Harel State Model*

6.1.1 This Standard uses the Harel state chart convention for describing dynamic operation of defined objects. The outline of this convention is described in an attachment of SEMI E30. The official definition of this convention is described in Science of Computer Programming 8, *Statecharts: A Visual Formalism for Complex Systems* by D. Harel, 1987.[1]

6.1.2 A transition table is used with the state chart for clearly describing the character of each state transition. The table contains the following: a transition number, current state, trigger, new state, and operation in transition.

6.1.3 The state models included in this Standard are a requirement for Substrate Tracking compliance. A state model consists of a state model diagram, state definitions, and a state transition table. When using collection events, all state transitions in this Standard, unless otherwise specified, shall correspond to collection events.

6.1.4 A state model represents the host's view of the equipment and does not necessarily describe the internal equipment operation. When using collection events, all Substrate Tracking state model transitions shall be mapped sequentially into the appropriate internal equipment collection events that satisfy the requirements of those transitions. In certain implementations, the equipment may enter a state and have already satisfied all of the conditions required by the Substrate Tracking state models for transition to another state. In this case, the equipment makes the required transition without any additional actions.

6.2 *OMT Object Information Model*

6.2.1 The object models are represented using the Object Modeling Technique (OMT) developed by Rumbaugh, James, *et al.*, in *Object Oriented Modeling Design*, 1991[2].

6.2.2 An overview of this notation is provided in SEMI E39, Appendix 1.

[1] Elsevier Science, P.O. Box 945, New York, NY 10159-0945; http://www.elsevier.nl/homepage/browse.htt

[2] Prentice Hall, Inc., Upper Saddle River, NJ 07458; http://www.prenhall.com/divisions/ecs/cscat.html

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6.3 *Object Attributes Representation*

6.3.1 The object information model for standardized objects will be supported by an attribute definition table with the following column headings:

Attribute Name	*Definition*	*Access*	*Reqmt*	*Format*
The formal text name of the attributes.	Description of information contained.	RO or RW	Y or N	Refer to the description below.

6.3.1.1 The Access column uses RO (read only) or RW (read/write) to indicate the access that a service user has to the attribute.

6.3.1.2 A 'Y' or 'N' in the Requirement (Reqmt) column indicates if this attribute must be supported in order to meet the fundamental requirement for the service.

6.3.1.3 The Format column is used for showing the data type of the attribute (refer to ¶ 5.2).

6.4 *Service Message Representation*

6.4.1 *Service Resource Definition* — A service resource definition table defines the specific sets of messages for a service group, as shown in the following table:

Message Service Name	*Type*	*Description*
Message name	N or R	Purpose of the service.

6.4.1.1 Type can be either N = Notification or R = Request. Notification type messages are initiated by a service provider and the provider does not expect to get the response from the service user or requester.

6.4.1.2 Request messages are initiated by a service user or requester. The Request message asks for data or an activity from the provider. Request messages expect a specific response message (no presumption on the message context).

6.4.2 *Service Parameter Dictionary* — A service parameter dictionary table defines the parameters for one or more services, as shown in the following table:

Parameter	*Form*	*Description*
Parameter X	Data type	The parameter called X is B in A.

6.4.2.1 A row is provided in the table for each parameter of the service. The first column contains the name of the parameter. This is followed by columns describing form and contents of the corresponding primitive.

6.4.2.2 The Form column is used to indicate the type of data contained in a parameter (refer to ¶ 5.2 for definitions).

6.4.2.3 The Description column in the Service Parameter Dictionary table describes the meaning of the parameter, the values it can assume, and any interrelationships with other parameters.

6.4.2.4 To prevent the definition of numerous parameters named 'XxxList', this Standard adopts the convention of referring to the list as '(List of) Xxxx'. In this case, the definition of the variable Xxx will be given, not of the list. The term 'list' indicates a collection (or set) of zero or more items of the same data type. Where a list is used in both the request and the response, the list order in the request is retained in the response. A list must contain at least one element unless zero elements are specifically allowed.

6.4.3 *Service Message Definition* — A service message definition table defines the parameter used in a service, as shown in the following table:

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Parameter	*Req/Ind*	*Rsp/Conf*	*Description*
Parameter X	see below	see below	A description of the service.

6.4.3.1 The columns labeled Req/Ind and Rsp/Conf link the parameters to the direction of the message. The message sent by the initiator is called the 'Request'. The receiver terms this message the 'Indication'. The receiver may then send a 'Response', which the original sender terms the 'Confirmation'.

6.4.3.2 The following codes appear in the Req/Ind and Rsp/Conf columns, and are used in the definition of the parameters (e.g., how each parameter is used in each direction):

M	Mandatory Parameter – must be given a valid value.
C	Conditional Parameter – may be defined in some circumstances and undefined in others. Whether a value is given may be completely optional or may depend on the values of other parameters.
U	User-Defined Parameter.
-	The parameter is not used.
=	(for response only) Indicates that the value of this parameter in the response must match that in the primary (if defined).

7 Overview

7.1 *Purpose of Substrate Tracking*

7.1.1 Substrate tracking consists of the following capabilities, as described below:

7.1.2 *Substrate Location Tracking* — Substrate Location Tracking provides the capability to determine the current location of substrates in the equipment. The indication of the substrate location will help the user to understand the environment to which the substrate is exposed.

7.1.3 *Batch Location Tracking* — Batch Location Tracking provides the capability to determine the current location of a group of substrates in the equipment. Typical example of a substrates group is a batch in batch process equipment.

7.1.4 *Substrate History Record* — Substrate History Record provides the capability to read the history of substrates in the equipment. The user can inquire about the history of a particular substrate. A history includes the series of locations that the substrate has visited.

7.1.5 *Substrate Process Tracking* — Substrate Process Tracking provides the capability of tracking current substrate processing state. For example, the user may determine whether a substrate is processed or not by requesting its state when the process has been completed abnormally. The user may use the state transitions to trigger wafer level data collection.

7.2 *View of Substrate Tracking*

7.2.1 Figure 1 shows the view of Substrate Tracking using the OMT representation. It is assumed that substrates are available before usage of the services. Substrate carriers may or may not be used to load the substrates to the equipment. Substrates are loaded into the equipment and travel through substrate locations and batch locations. The equipment must maintain all information necessary to track substrates in the equipment.

7.3 *Substrate Location and Batch Location*

7.3.1 A substrate location is a discrete position that can hold a substrate in the equipment. This Standard will not define where the substrate location exists. Substrate locations shall be determined by the design of equipment and the applications. For example, to track the order of process chambers that is applied to a substrate process in a multi-chamber single wafer process equipment, it is appropriate to designate process chambers for substrate locations. On the other hand, in batch process equipment, process chambers are designated for batch locations and substrate locations are not used to track the order of process chambers that is applied to the substrates of a batch. All substrate locations shall be documented by the equipment supplier.

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7.3.2 Batch Location indicates the physical location of a group of substrates (e.g., batch) on the equipment. Generally, in batch process equipment, Batch Location is assigned to a process chamber and changed when a batch is transferred from one chamber to another. Individual substrate positions in a batch (e.g., slot positions of a batch container or a process carrier) might be informed with Batch Location. Batch Locations and notation of the substrate positions in a batch are defined and documented by the equipment supplier.

7.3.3 Figure 2 shows the concept of the substrate location using OMT representation. This figure is provided to clarify the definition of locations.

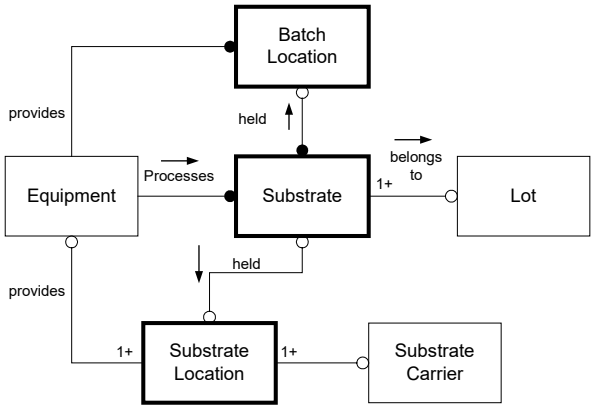

Figure 1
Overview of Substrate Tracking

7.3.4 Material Location is classified into Substrate Location, Batch Location, and Carrier Location. A Carrier Location may hold a carrier, a Substrate Location may hold a substrate, and a Batch Location may hold a group of substrates. Typical example of a substrates group is a batch in batch process equipment.

7.3.5 Substrate Location is classified into Equipment Substrate Location and Carrier Substrate Location. Carrier has Carrier Substrate Locations. A Carrier Substrate Location is the location which can hold a substrate in a carrier. An Equipment Substrate Location is the location which can hold a substrate on the equipment resource.

7.3.6 Substrate travels on substrate locations and/or batch locations.

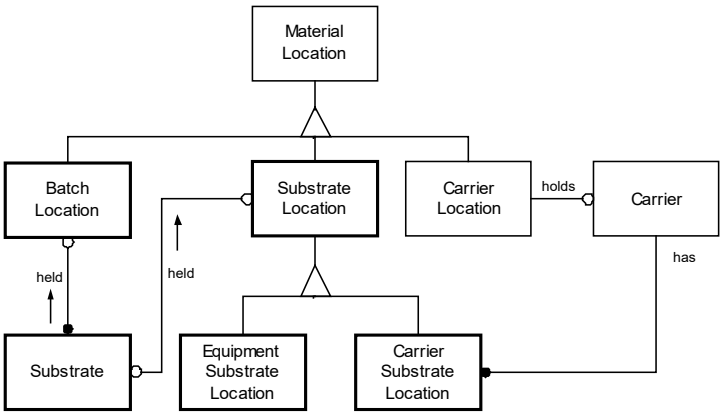

Figure 2
Concept of Substrate Location

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7.4 *Identification of Substrate*

7.4.1 Substrate shall be identified by substrate ID. There are options for assigning substrate ID to a substrate.

- When a carrier is used for loading substrates and the carrier is given its carrier ID, substrate IDs can be obtained by utilizing the carrier ID and carrier substrate location. This Standard will define the requirement for the substrate ID assignment to the substrates in a carrier.
- Substrate ID may be read from the substrate directly by a substrate ID reader. The substrate ID read from the substrate can be used for the identification.
- Substrate ID may be assigned by the user (i.e., host or operator). The substrate ID can be used for the identification. This Standard will define the service for registering substrates.

7.4.2 The recommendation of a particular option for assigning substrate IDs is not within the scope of this Standard. It must be selected based on the equipment design and applications.

8 Substrate Tracking Requirement

8.1 Compliance to Substrate Tracking Service (STS) requires that an implementation also provide certain capabilities that are defined in other standards.

- STS requires compliance to SEMI E39 (OSS) for Substrate objects. The attributes of these objects may be accessed through use of the GetAttr and SetAttr services.
- STS requires an event reporting capability. This capability may be satisfied by the Event Reporting capability in SEMI E30 (GEM).

8.2 STS compliant equipment must implement capabilities defined in the following sections of this Standard:

- § 9, Substrate Object Definition,
- § 10, Substrate Location Object Definition,
- § 12, Substrate Tracking Services,
- § 13, Variable Data, and
- § 11, Batch Location Object Definition, is an optional requirement.

8.3 When the equipment provides a substrate reader, the equipment shall implement an equipment constant to allow the host to choose between using or not using this capability. The equipment constant: ‘SubstrateReaderEnabled’ is used to choose whether or not the equipment performs the reading of the substrate and use the Substrate Reading Status state model. The equipment is responsible for creating the substrate objects. The substrate ID assigned to the substrate object is the same as the one provided by the host in the ContentMap attribute or created by the equipment using the following rule when the SubstID in the content map is not provided. The default substrate ID shall be represented in text as the carrierID, a period and the substrate slot location. For example, if CarrierID = ‘xyz’ and substrate slot location = ‘5’, then the substrate ID would be = ‘xyz.05’.

8.4 All state transitions defined for the three state models specified in this Standard must be reportable via discrete collection events as defined in ¶ 6.1, Harel State Model. The three state models are the Substrate Object State Model (Figure 3), the Substrate Location State Model (Figure 4) and the Batch Location State Model (Figure 5).

8.4.1 The data in Table 14 (Variables Required for Substrate Object State Model Transitions) is required to be available for the Substrate Object State Model transition collection events. Fixed buffer equipment must support discrete state transitions for Substrate. Internal buffer equipment must support related state transitions that occur for a group of Substrate objects. An equipment may support both discrete and related state transitions for Substrate.

8.4.2 The data in Table 15 (Variables Required for Substrate Location Object State Model Transitions) is required to be available for the Substrate Location Object State Model transition collection events.

8.4.3 The data in Table 16 (Variables Required for Batch Location Object State Model Transitions) is required to be available for the Batch Location Object State Model transition collection events.

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8.5 Each Substrate Object instantiated on the equipment must maintain an independent instance of the Substrate Object State Model. When discrete and unrelated state transitions occur for Substrate Object models, each shall trigger a separate collection event. However, when related state transitions occur for a group of Substrate Object models (such as with internal buffer equipment material transfers), each model does not trigger a separate collection event. Instead, a single collection event shall be triggered (for all related Substrate Objects in the group). The data items defined in Table 14 are required to be valid for collection events triggered by state model transitions in the following way:

8.5.1 These variables from Table 14 shall be valid for collection events triggered by discrete state transitions:

- AcquiredID (equipment with substrate ID readers only)
- AdditionalInfo (if supported)
- SubstID
- SubstIDStatus (equipment with substrate ID readers only)
- SubstBatchLocID (batch process equipment only)
- SubstDestination
- SubstHistory
- SubstSubstLocID
- SubstLotID
- SubstMtrlStatus
- SubstObjType
- SubstPosInBatch (batch process equipment only)
- SubstProcState
- SubstSource
- SubstState
- SubstType
- SubstUsage

8.5.2 These variables from Table 14 shall be valid for collection events triggered by a group of related state transitions:

- AcquiredIDList (equipment with substrate ID readers only)
- AdditionalInfoList (if supported)
- SubstIDList
- SubstIDStatusList (equipment with substrate ID readers only)
- SubstBatchLocIDList (batch process equipment only)
- SubstDestinationList
- SubstHistoryList
- SubstLocIDList
- SubstLotIDList
- SubstMtrlStatusList
- SubstObjTypeList
- SubstPosInBatchList (batch process equipment only)

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- SubstProcStateList
- SubstSourceList
- SubstStateList
- SubstTypeList
- SubstUsageList

8.5.3 As an example of how the variables are constructed for related state transitions, the value of SubstIDList shall be the list of SubstID data items that identify the Substrate Objects related by the collection event.

8.6 Each Substrate Location Object instantiated on the equipment must maintain an independent instance of the appropriate Substrate Location Object State Model. Each Substrate Location Object shall trigger a separate collection event for each state transition. The data items defined in Table 15 are required to be valid for collection events triggered by each state model transition.

8.7 Each Batch Location Object instantiated on the equipment must maintain an independent instance of the Batch Location Object State Model. Each Batch Location Object shall trigger a separate collection event for each state transition. The data items defined in Table 16 are required to be valid for collection events triggered by each state model transition.

9 Substrate Object Definition

9.1 A substrate is a base upon which a product unit is built in the manufacturing equipment. Wafers and flat panel displays are examples of substrates. A substrate may be associated with a lot, which the factory uses for tracking purposes.

9.1.1 The substrate object shall be created when the substrate is registered by the equipment. There are two possible ways of registering the substrate.

9.1.2 The substrate is registered by the equipment when the information for the carrier containing the substrate is registered. The information required to create a substrate object must be provided with the carrier. However, the specification of carrier management services is outside the scope of this Standard. Substrate Tracking implementations shall be consistent with the Carrier Management Standard. For instance, if the Carrier Management Standard specifies the method of identifying the substrates in a carrier, then Substrate Tracking shall recognize those identifiers.

9.1.3 The substrate is registered when the substrate is transferred directly (not in a carrier) to the equipment by the Create service defined in SEMI E39 to inform the equipment of this event.

9.2 *Substrate Object State Model*

9.2.1 A substrate has both a state indicating where it is located and a state indicating the progress of processing. These states are represented by the Substrate Transport and the Substrate Processing states, which are concurrent substates of the Substrate State Model shown in Figure 3.

9.2.2 If the equipment provides a mechanism for reading substrates, then, the equipment shall include a state model in parallel with the Substrate Transport and Substrate Processing States called ‘Substrate Reading Status’ see Figure 3 and Table 1 for additional requirements.

9.3 *Substrate Object State Definitions*

9.3.1 *ABORTED* — The process has been aborted during the processing. The substrate will require special treatment.

9.3.2 *AT DESTINATION* — The substrate has been placed at its designated destination substrate location. The substrate may have been placed at a carrier substrate location when the carrier is used to unload the substrate from the equipment or at an equipment substrate location when the substrate is to be removed by itself from the equipment.

9.3.3 *AT SOURCE* — The state which the substrate is originally received. The substrate is held at carrier substrate location when a carrier is used to supply the substrate.

9.3.4 *AT WORK* — The substrate has been taken from the original substrate location or destination substrate location, and is traveling on intermediate equipment substrate locations. The substrate has been taken out of the carrier and placed into the equipment when a carrier is used to supply the substrate.

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9.3.5 *IN PROCESS* — Substrate is being processed. One or more substrate properties are being changed by the processing instructions or measurement of the substrate in the equipment. The nature of the process depends upon the equipment processing capabilities.

9.3.6 *LOST* — The terminal state when the substrate has been broken and/or removed from the equipment by an external entity and it no longer exists at the equipment.

9.3.7 *NEEDS PROCESSING* — Processing requirements exist that have not yet been fulfilled. This is the default entry state when the substrate is originally received. In some cases, the substrate may return to this state while it waits for additional processing to be performed.

9.3.8 *PROCESSED* — All substrate processing has successfully completed. No further processing will be performed by the equipment.

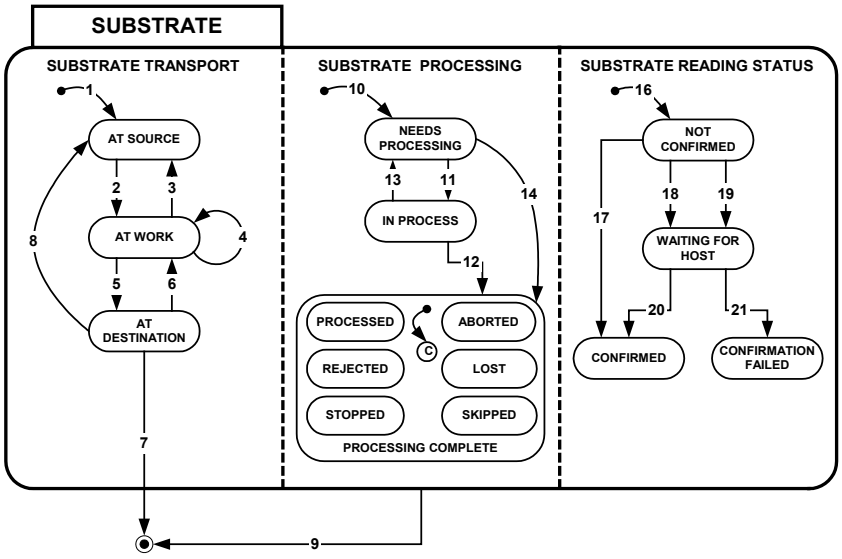

Figure 3
Substrate Object State Model

9.3.9 *PROCESSING COMPLETE* — The state in which the substrate has completed all processing associated with a recipe specification. The substrate processing completed because the process requirement was aborted, stopped or successfully completed. The substrate is no longer changed by the recipe's processing instructions or its measurement/inspection requirements completed. This state is a super-state of the terminal states of the substrate; PROCESSED, ABORTED, REJECTED, LOST, STOPPED, or SKIPPED.

9.3.10 *REJECTED* — The state which the substrate has been processed completely; however, the result of the processing may have a problem. The substrate will require special treatment. This substrate is not required for equipment that is unable to identify rejected substrates.

9.3.11 *SKIPPED* — As directed by an operator or a host, the substrate was not processed.

9.3.12 *STOPPED* — The process has been stopped during the processing. The substrate will require special treatment.

9.3.13 *SUBSTRATE* — The superstate of the SUBSTRATE TRANSPORT and SUBSTRATE PROCESSING. SUBSTRATE TRANSPORT and SUBSTRATE PROCESSING are concurrent substrates. The state is created when the substrate is initially registered to the equipment, and deleted when the substrate is removed from the equipment.

9.3.14 *SUBSTRATE PROCESSING* — The superstate of NEEDS PROCESSING, IN PROCESS, PROCESSING COMPLETE which represents the processing state of the substrate.

9.3.15 *SUBSTRATE TRANSPORT* — The superstate of the AT SOURCE, AT WORK, AT DESTINATION which represents the transport state of the substrate within the equipment.

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9.3.16 *SUBSTRATE READING STATUS* — The superstate of NOT CONFIRMED, WAITING FOR HOST, CONFIRMED and CONFIRMATION FAILED that represents the status of the substrate ID confirmation by the equipment.

9.3.17 *NOT CONFIRMED* — This is the entry state for the substrate object at the time of its creation when a substrate reader is available and enabled in the equipment.

9.3.18 *WAITING FOR HOST* — The status of the substrate when the substrate ID could not be identified by the reader, was discovered to be at a wrong source location or there was no matching ID in the source set.

9.3.19 *CONFIRMED* — The state of the substrate when its substrate ID has been confirmed by the equipment or the host after it was read.

9.3.20 *CONFIRMATION FAILED* — The state of the substrate when the host decides to cancel the substrate.

9.4 *Substrate State Transition*

Table 1 Substrate State Model Transition Table

No.	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	no state	The substrate is registered.	AT SOURCE	A new record is created and added to SubstHistory. The new record shall have its values initialized as follows: 1) TimeIn is an actual value. 2) TimeOut is an empty value. 3) location is initialized in one of the following ways: *a) If the substrate is instantiated in a Substrate Location:* location is equal to the SubstLocID of the Substrate Location where the Substrate was instantiated. *b) If the substrate is instantiated into a Batch Location:* location is equal to BatchLocID+“.”+SubstPosIn Batch. Note: BatchLocID corresponds to the Batch Location where the Substrate was instantiated and SubstPosInBatch corresponds to the position of the Substrate whose SubstHistory is being updated.	Substrate tracking begins.
2	AT SOURCE	The substrate is taken from the source substrate location and placed into the equipment.	AT WORK		

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No.	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
3	AT WORK	The substrate has moved to the source substrate location.	AT SOURCE		This transition is not required for compliance to this Standard.
4	AT WORK	The substrate has moved from an equipment material location to a different equipment material location. Note: The material location can be either an Equipment Substrate or a Batch Location.	AT WORK		This transition is not required if the equipment only provides a single Equipment Substrate Location or a single Batch Location.
5	AT WORK	The substrate is moved to the destination substrate location or batch location.	AT DESTINATION		
6	AT DESTINATION	The substrate is taken from the destination substrate location and placed into the equipment.	AT WORK		This transition is not required for compliance to this Standard.
7	AT DESTINATION	The substrate is removed from the equipment by a normal transfer sequence.	(Extinction)	The substrate object is deleted in the equipment.	
8	AT DESTINATION	The user informs or the equipment detects that the substrate is AT SOURCE.	AT SOURCE		This transition is not required for compliance to this Standard.
9	Any SUBSTRATE substate	The equipment detects or is informed by the user that a substrate has been removed.	(Extinction)	The substrate object is deleted in the equipment.	The removal may be detected by the equipment or informed by the user. This transition would not occur during normal wafer movement. So, if transition 7 occurs, there is no need to report a transition 9.
10	no state	The substrate object is created.	NEEDS PROCESSING		

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No.	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
11	NEEDS PROCESSING	Substrate starts actual processing. One or more substrate properties begin to be affected by the equipment recipe instructions or measurement of the substrate.	IN PROCESS		The document for each equipment implementation shall describe the trigger for this transition.
12	IN PROCESS	The processing completes because the processing activities were aborted, stopped or completed. One or more substrate properties are no longer affected by the processing instructions or measurement of the substrate.	PROCESSING COMPLETE	One of the substates PROCESSED, ABORTED, STOPPED, REJECTED, LOST, SKIPPED is determined.	This is a terminal state. The substrate either returns to the specified destination or it is no longer at the equipment.
13	IN PROCESS	The process associated with the processing instructions completed and the substrate is to be processed again.	NEEDS PROCESSING		The documentation for each equipment implementation shall describe the trigger for this transition. This transition is not required for compliance to this Standard.
14	NEED PROCESSING	The substrate has been removed from the equipment by an external agent or it is physically missing (LOST) or the substrate has been not processed (SKIPPED).	PROCESSING COMPLETE	One of the substrates LOST, SKIPPED, is determined.	
16	no state	The substrate object is created.	NOT CONFIRMED	None.	This transition is not reported.
17	NOT CONFIRMED	Substrate was successfully read and SubstID provided in the ContentMap matched the AcquiredID read by the equipment.	CONFIRMED	Issue an event for transition 17.	Data required to be available for this event report: SubstID AcquiredID
18	NOT CONFIRMED	Substrate ID reading has failed after several retries.	WAITING FOR HOST	Issue an event for transition 18. Wait for the host to issue either ‘ProceedWithSubstrate’ or ‘CancelSubstrate’ services.	Data required to be available for this event report: SubstID

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No.	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
19	NOT CONFIRMED	Substrate ID was successfully read but the acquired ID is different from the one the equipment used to instantiate the substrate object.	WAITING FOR HOST	Issue an event for transition 19. Wait for the host to issue either 'ProceedWithSubstrate' or 'CancelSubstrate' services.	Data required to be available for this event report: SubstID AcquiredID
20	WAITING FOR HOST	Equipment has received a 'ProceedWith Substrate' service.	CONFIRMED	Issue an event for transition 20.	Data required to be available for this event report: SubstID AcquiredID (if valid)
21	WAITING FOR HOST	Equipment has received a 'CancelSubstrate' service.	CONFIRMATION FAILED	Issue an event for transition 21 and update the substrate processing status to SKIPPED taking the appropriate transition. The substrate is returned or placed into its destination location.	

9.5 *Substrate Object Attribute Definition*

9.5.1 Table 2 defines attributes for substrate objects. These attributes can be accessed by using GetAttr and SetAttr messages as defined in SEMI E39 (OSS).

Table 2 Substrate Object Attribute Table

Attribute Name	*Definition*	*Access*	*Reqmt*	*Format*
AcquiredID	Contains the ID read from the substrate. Empty string before the substrate is read. The attribute shall be updated as soon the substrate ID has been successfully read. [Equipment with substrate ID reader only]	RO	N	Text
AdditionalInfo	Specifies additional information for the substrate through name-value pairs. Refer to § 9.5.3.	RO	N	List of structures consisting of two fields {name, value}
BatchLocID	Identifier for current batch location.	RO	N	Text (batch process equipment only)
LotID	Identifier of the lot associated with the substrate by the user.	RW	N	Text
MaterialStatus	Current status of the substrate that represents criteria of the processing quality. The criteria is equipment design dependent.	RO	N	Enumerated: equipment dependent
ObjID	Object identifier.	RO	Y	Text equal to the Substrate ID
ObjType	Object type.	RO	Y	Text = 'Substrate'
SubstDestination	Identifier of the substrate location on which the substrate shall be finally restored. When a carrier is used for restoring the substrate, the value shall be the concatenation of carrier identifier (carrier ID) string and a numeric string representing the substrate position in the carrier to represent the carrier and position of the substrate restored.	RO	Y	Text If empty string, then same as source substrate location

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Attribute Name	*Definition*	*Access*	*Reqmt*	*Format*
SubstHistory	History of locations visited.	RO	Y	List of structures consisting of SubstLocID or BatchLocID+“.”+SubstPosInBatch (batch process equipment only), TimeIn, TimeOut
SubstIDStatus	Status of the substrate ID. This attribute is mandatory only when the equipment provided a substrate reader.	RO	N	Enumerated NOT CONFIRMED WAITING FOR HOST CONFIRMED CONFIRMATION FAILED
SubstLocID	Identifier for current equipment substrate location or carrier substrate location that the substrate occupies.	RO	Y	Text (If batch process equipment, this data is empty while the substrate belongs to a batch)
SubstPosInBatch	Current substrate position in a batch.	RO	N	Text (batch process equipment only, and empty if substrate position in a batch is not informed)
SubstProcState	Processing state of the substrate. NEEDS PROCESSING: initial value to indicate that processing in the equipment is required and has not started. IN PROCESS: processing has started. PROCESSED: processing has completed normally. ABORTED: processing has terminated abnormally. STOPPED: processing has terminated at completion of certain processing step. REJECTED: the substrate has been identified as needing special treatment. LOST: the substrate has been removed from the equipment by an external agent or it is physically missing. SKIPPED: the substrate was not processed.	RO	Y	Enumerated: NEEDS PROCESSING IN PROCESS PROCESSED ABORTED STOPPED REJECTED LOST SKIPPED
SubstSource	Identifier of the substrate location on which the substrate has been initially registered. When a carrier is used for registering the substrate, the value shall be the concatenation of carrier identifier (carrier ID) string and a numeric string representing the substrate position in the carrier to represent the carrier and position of the substrate registered.	RO	Y	Text
SubstState	Transport state of the substrate.	RO	Y	Enumerated: AT SOURCE AT WORK AT DESTINATION
SubstType	The type of the substrate. It includes wafers, flat panels, CD’s or masks.	RW	N	Enumerated: WAFER FLAT PANEL CD MASK
SubstUsage	How the substrate is used, such as ‘product’, ‘test’, and ‘filler’.	RW	N	Enumerated: PRODUCT TEST FILLER Additional types defined and added by the equipment.

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9.5.2 SubstHistory attribute is a composite data type. The constitute data is defined in Table 3.

9.5.3 *AdditionalInfo Attribute* — This attribute provides a generic mechanism to provide additional information about the substrate through name-value pairs, rather than creating custom attributes on the SEMI E90 substrate object.

9.5.3.1 This attribute can be used to provide custom information about the substrate to the equipment, and/or provide equipment-generated information about the substrate. For example, it could be used to specify the substrate thickness information or report what carrier the substrate originated from.

9.5.3.2 Supported AdditionalInfo element names are specified by the equipment.

NOTE 1: The equipment and host negotiate how data in AddtionalInfo attribute is to be used. Such negotiations are outside the scope of this Specification.

9.5.3.3 Information already defined as SEMI E90 substrate attributes are not accessed through the AdditionalInfo attribute. For example, LotID and SubstUsage information are already defined as substrate attributes and are not specified as elements in the carrier's AdditionalInfo attribute.

9.5.3.4 The Equipment could add or delete elements in the AdditionalInfo attribute, or it could change the value for an existing name-value pair.

9.5.3.5 The host cannot change the AdditionalInfo attribute after the substrate has been created.

NOTE 2: The AdditionalInfo attribute is read-only because if the Host changes the value, there is an expectation of some corresponding behavior change on the equipment. Since the name-value pair is implementation specific, the Specification can't dictate what the behavior change will be.

Table 3 Attribute Data Definition

Data Identifier	*Description*	*Form*
BatchLocID+"."+ SubstPosInBatch	Batch location, on which the group of substrates (e.g., batch) has visited, with substrate position information in a batch (recorded while the substrate belongs to a batch). For example, if BatchLocID = 'Chamber-A' and SubstPosInBatch = '30', then what is recorded is 'Chamber-A.30'.	Text (batch process equipment only, and if substrate position in a batch is not informed "."+SubstPosInBatch portion is not recorded).
SubstLocID	Equipment substrate location or carrier substrate location on which the substrate was registered or has visited. Must be copied from SubstLocID of which the substrate was registered or has visited. (For batch process equipment, this data is not recorded while the substrate belongs to a batch.)	Text.
TimeIn	Actual arrival time of the substrate on the substrate location.	The format of the timestamp defined by the protocol for implementation.
TimeOut	Actual departure time of the substrate from the substrate location or batch location. This will be an 'empty' item until the substrate has departed the location.	The format of the timestamp defined by the protocol for implementation.

10 Substrate Location Object Definition

10.1 A substrate location object (SLO) provides a model for identifying substrate locations. Each SLO on an equipment is assigned a substrate location ID to uniquely identify it. The assignment shall be documented by the equipment supplier. There are two types of substrate locations: carrier substrate location, which is the location or position (e.g., slot) in the carrier, and equipment substrate location, which is on the equipment resource. The equipment substrate location is a persistent object, while the carrier substrate locations are dynamic objects that shall be created or deleted by the placement or removal of carriers on the equipment.

10.1.1 Source substrate locations and Destination substrate locations are the points at which substrates transfer to/from the equipment's internal substrate locations (often locations at which processing occurs). A carrier substrate location is the Source or Destination substrate location when a carrier is used to transfer the substrate. An equipment

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substrate location can be the Source or Destination substrate location when the substrate is transferred directly (without a carrier).

10.2 *Substrate Location State Model*

10.2.1 Figure 4 shows the dynamic behavior of the substrate location using the Harel state chart representation.

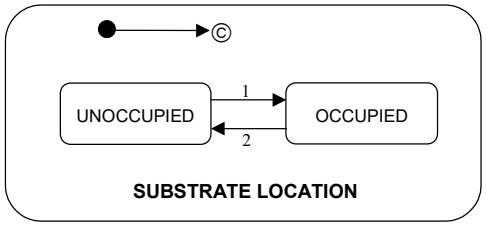

Figure 4
Dynamic Behavior Model of Substrate Location

10.3 *Substrate Location State Definitions*

10.3.1 *SUBSTRATE LOCATION* — The superstate of UNOCCUPIED and OCCUPIED.

10.3.2 *UNOCCUPIED* — The state in which the substrate location does not hold or have a substrate.

10.3.3 *OCCUPIED* — The state in which the substrate location holds a substrate.

10.4 *Substrate Location State Transition*

Table 4 Substrate Location State Model Transition Table

No.	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	UNOCCUPIED	Substrate moves onto the substrate location.	OCCUPIED	Update the SubstHistory attribute of the substrate that arrived (or was registered) as follows: A new record is created and added to SubstHistory. TimeIn is an actual value, TimeOut is an empty value, location is equal to the SubstLocID of the Substrate Location that underwent this transition.	
2	OCCUPIED	Substrate moves off the substrate location.	UNOCCUPIED	Update the SubstHistory attribute of the substrate that departed by populating the TimeOut element of the record that was created when the substrate arrived in the Substrate Location.	

10.5 *Substrate Location Object Attributes*

10.5.1 Table 5 defines attributes for substrate location objects. These attributes can be accessed by using GetAttr and SetAttr messages as defined in SEMI E39 (OSS).

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Table 5 Substrate Location Object Attribute Table

Attribute Name	*Definition*	*Access*	*Reqmt*	*Format*
DisableEvents	When set to TRUE, the transitions for this substrate location do not generate events to the host.	RW	Y	Boolean TRUE = Disable FALSE = Enable
ObjID	Object Identifier.	RO	Y	Text equal to the Substrate Location ID
ObjType	Object type.	RO	Y	Text = 'SubstLoc'
SubstID	Substrate Identifier relevant to the location.	RO	Y	Text
SubstLocState	Substrate Location state.	RO	Y	Enumerated: UNOCCUPIED, OCCUPIED

11 Batch Location Object

11.1 A batch location object (BLO) provides a model for identifying a group of substrates (e.g., batch) locations. Each BLO on an equipment is assigned a batch location ID to uniquely identify it. The assignment shall be documented by the equipment supplier. The batch location is a persistent object.

11.2 *Batch Location State Model*

11.2.1 Figure 5 shows the dynamic behavior of the batch location using the Harel state chart representation.

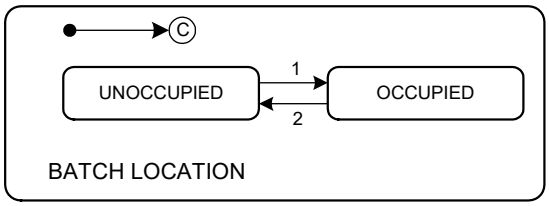

Figure 5
Dynamic Behavior Model of Batch Location

11.3 *Batch Location State Definitions*

11.3.1 *BATCH LOCATION* — The superstate of UNOCCUPIED and OCCUPIED.

11.3.2 *UNOCCUPIED* — The state in which the batch location does not hold or have any substrate of batch.

11.3.3 *OCCUPIED* — The state in which the batch location holds one or more substrates of batch.

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11.4 *Batch Location State Transition*

Table 6 Batch Location State Model Transition Table

No.	*Current State*	*Trigger*	*New State*	*Action*	*Comment*
1	UNOCCUPIED	All substrates of the group (e.g., batch) have moved onto the batch location.	OCCUPIED	Update the SubstHistory attribute for each substrate that arrived (or was registered) as follows: A new record is created and added to SubstHistory. TimeIn is an actual value, TimeOut is an empty value, location is equal to BatchLocID+“.”+SubstPosIn Batch. Note: BatchLocID corresponds to the Batch Location that underwent this state transistion and SubstPosInBatch corresponds to the position of the substrate whose SubstHistory is being updated.	No events will be reported during subsequent creation of a batch.
2	OCCUPIED	All substrates of the group (e.g., batch) moves off the location.	UNOCCUPIED	Update the SubstHistory attribute for each substrate that departed as follows: Populate the TimeOut element of the record that was created when the substrate arrived in the Batch Location.	No events will be reported during subsequent destruction of a batch.

11.5 *Batch Location Object Attributes*

11.5.1 Table 7 defines attributes for batch location objects. These attributes can be accessed by using GetAttr and SetAttr messages as defined in SEMI E39 (OSS).

Table 7 Batch Location Object Attribute Table

Attribute Name	*Definition*	*Access*	*Reqmt*	*Format*
ObjID	Object Identifier.	RO	Y	Text equal to the Batch Location ID
ObjType	Object type.	RO	Y	Text = ‘BatchLoc’
BatchLocState	Batch Location state.	RO	Y	Enumerated: UNOCCUPIED, OCCUPIED
BatchSubstIDMap	Ordered list of substrate identifiers corresponding to the positions in a batch (e.g., slot positions of a batch container or a process carrier).	RO	Y	Ordered list of SubstID (Ordered list indicates position in the batch and SubstID can be actual substrate identifier, blank “” when location is empty or ‘filler’ when such substrate is used.) This attribute is updated each time a substrate is moved into or out of the batch or as soon as the equipment has the information.

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Attribute Name	*Definition*	*Access*	*Reqmt*	*Format*
DisableEvents	When set to TRUE, the transitions for this batch location do not generate events to the host.	RW	Y	Boolean TRUE = Disable FALSE = Enable

12 Substrate Tracking Services

12.1 *Creating and Deleting Substrate Objects*

12.1.1 Equipment may create substrate objects as appropriate during equipment operation. On some equipment (at the option of the equipment designer), substrate objects may also be registered (created) at the equipment through the host-initiated Create service defined in SEMI E39. Equipment is not required to provide host-initiated Register (Create) Substrate or Remove Substrate services for fundamental compliance to STS. Table 8 defines the attribute settings that may be initialized through this service.

12.1.1.1 If the equipment does not support the AdditionalInfo attribute and the Create service specifies the AdditionalInfo attribute as anything other than an empty list, the Create service shall fail.

12.1.1.2 Only valid element names supported by the equipment for the AdditionalInfo attribute elements can be specified in the AdditionalInfo attribute. If the Create service specifies the AdditionalInfo attribute with an invalid element name, the Create service shall fail.

12.1.2 Each application must specify its own requirements regarding which optional attributes settings will be provided in the Create service.

12.1.3 The Create service always requires ObjType to be specified as an argument. The Create service also allows for setting of attribute values (as arguments separate from ObjType). The ObjType attribute is present in all SEMI defined objects. Therefore, the create service potentially allows the sender to specify ObjType twice, once as the ObjType argument and again as an attribute setting for ObjType. In order to prevent ambiguity (e.g., the sender specified the ObjType argument as one value and the attribute setting for ObjType as a different value) the sender is restricted from sending an attribute setting for ObjType and the equipment shall ignore an attribute setting for ObjType if provided. In the table, M indicates Mandatory, O indicates Optional, and C represents Conditional (dependent on equipment structure and the location where the Substrate object is to be instantiated). The Conditional attribute settings SubstLocID, BatchLocID and SubstPosInBatch shall be provided as follows: If instantiating a Substrate object into a Substrate Location, SubstLocID shall be provided but BatchLocID and SubstPosInBatch shall not be provided. If instantiating a Substrate object into a Batch Location then BatchLocID and SubstPosInBatch shall be provided but SubstLocID shall not be provided. It is not permitted to use the host-initiated Create service without providing the location because the location allows the equipment to identify the corresponding physical substrate unambiguously.

Table 8 Substrate Attributes Settable through Create

Substrate Attribute Name	*Use as AttrSetting in Create Service*
AdditionalInfo	O
BatchLocID	C (batch process equipment only)
LotID	O
MaterialStatus	O
ObjID	M
SubstDestination	O
SubstHistory	O
SubstLocID	C
SubstProcState	O
SubstPosInBatch	C (batch process equipment only)
SubstSource	O

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Substrate Attribute Name	*Use as AttrSetting in Create Service*
SubstState	O
SubstType	O
SubstUsage	O

12.1.4 Use of attribute settings for the Delete service are optional and specific to the application.

12.2 Table 9 defines services required for substrate tracking. These services must be supported with substrate object.

Table 9 Substrate Tracking Service Message Definition Table

Message Service Name	*Type*	*Description*
CancelSubstrate	R	Request to the equipment to skip the substrate and move into its source location or destination location. Equipment instantiated substrate ID is required by this service. This service is only required when the equipment provides capabilities for reading substrate identifiers.
ProceedWithSubstrate	R	Request to the equipment to accept the substrate. Equipment instantiated substrate ID is required by this service. This service is only required when the equipment provides capabilities for reading substrate identifiers.
UpdateSubstrateObject	R	Request to create a substrate object or to continue updating existent substrate object attributes for the specified substrate identifier with the substrate information provided with the service. If the substrate object was previously instantiated, the attribute(s) of the substrate are updated as specified with this service. If transfer is not completed or substrate is not at the equipment the service request fails. This service is used when substrates are moved directly between linked equipment without the use of a carrier. This is an optional service.

12.3 *Substrate Tracking Service Parameter Dictionary*

Table 10 Substrate Tracking Service Parameter

Dictionary Parameter	*Form*	*Description*
ErrorCode	Enumerated	Contains the code for the specific error found.
ErrorText	Text	Text in support of the error code.
STAcknowledge	Enumerated	Acknowledge of request.
Status	A list of ErrorCode/ErrorText pairs	Report any errors found.
STStatus	A structure consisting of STAcknowledge and Status	Return information for a service.
SubstID	Text	Identifier of a substrate.
SubstLocID	Text	Identifier of substrate location.
SubProcState	Enumerated	Processing State of a substrate.
SubstState	Enumerated	Transport State of a substrate.

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Dictionary Parameter	*Form*	*Description*
(List of) SubstrateObjectAttributes	List of structure: Attribute name Attribute Value	List of substrate attributes. Each substrate attributes is an attribute name/value pair. The UpdateSubstrateObject service requires some substrate object attributes to be specified. For example, SubstrateID, SubstType, and at least one of the following must be included: SubstState, SubstProcState. All other substrate object properties are optional as is in the case of LotID, SubstSource, SubstUsage, SubstHistory, etc. Supplier may define attributes to be included in the service that are equipment dependent as in the case of linked litho tools.
ObjSpec	Refer to SEMI E39.	Specifier of the object owner.
ObjectActionStatus	Structure ServiceAcknowledge Status	
(List of) ObjectActionParameterResult	List of structure: Attribute name Attribute Value	List of return parameters regarding the result of the request as name/value pairs.
ObjectAction	Text	'UpdateSubstrateObject'
OperationID	Unsigned Integer	Identifies a specific request. Refer to SEMI E39.
ObjectLinkID	Unsigned Integer	Set to nonzero if and only if additional completion reports will be sent.
ServiceAcknowledge	Enumerated	0 – Acknowledge performed successful 1 – Service does not exist 2 – Cannot perform now 3 – At least one parameter is invalid 4 – Service will be performed later with notification 5 – Service is not completed or prohibited 6 – Not used 7–63 – Reserved
Status	List of structure Errorcode Errortext	

Table 11 UpdateSubstrateObject Service Parameter

Parameter	*Req/Ind*	*Rsp/Conf*	*Description*
ObjectAction	M	-	'UpdateSubstrateObject'
ObjSpec	C	-	The object specifier of the object owner. Refer to SEMI E39.
SubstrateObjectAttributes	M	=	SubstrateID, SubstType, and at least one of the following must be included: SubstState, SubstProcState. Refer to UpdateSubstrateObject service message definition for additional requirements.
OperationID	C	-	Used if multiple requests will be linked to events. Set to zero otherwise.
ObjectLinkID	-	C	Set to nonzero if additional completion reports will be sent.

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Parameter	*Req/Ind*	*Rsp/Conf*	*Description*
ObjectActionStatus	-	M	Acknowledge of the service acceptance and a list of error code and error text pairs if any.
ObjectActionParameterResult	-	C	List of return parameters regarding the result of the request as name/value pairs. Service returns the substrate object attributes updated that the request specified.

Table 12 ProceedWithSubstrate Service Parameter

Parameter	*Req/Ind*	*Rsp/Conf*	*Description*
ObjectAction	M	-	'ProceedWithSubstrate'
ObjSpec	M	-	The object specifier of the object owner. Refer to SEMI E39.
SubstrateObjectAttributes	M	=	SubstID, and SubstLocID are required.
OperationID	M	-	Used if multiple requests will be linked events. Set to zero otherwise.
ObjectLinkID	-	M	Set to nonzero if additional completion reports will be sent.
ObjectActionStatus	-	M	Acknowledge of the service acceptance and a list of error code and error text pairs if any.
ObjectActionParameterResult	-	M	SubstID, and SubstLocID are required.

Table 13 CancelSubstrate Service Parameter

Parameter	*Req/Ind*	*Rsp/Conf*	*Description*
ObjectAction	M	-	'CancelSubstrate'
ObjSpec	M	-	The object specifier of the object owner. Refer to SEMI E39.
SubstrateObjectAttributes	M	=	SubstID, and SubstLocID are required.
OperationID	M	-	Used if multiple requests will be linked events. Set to zero otherwise.
ObjectLinkID	-	M	Set to nonzero if additional completion reports will be sent.
ObjectActionStatus	-	M	Acknowledge of the service acceptance and a list of error code and error text pairs if any.
ObjectActionParameterResult	-	M	SubstID, and SubstLocID are required.

13 Variable Data

13.1 The host may inquire about any data related to specific Substrates, Substrate Locations and Batch Locations at any time by using the OSS service GetAttr.

13.2 Implementations of Specification for Substrate Tracking that uses the Event Reporting capability specified by SEMI E30 (GEM) have the following requirement: For the objects defined by Specification for Substrate Tracking, the tables listed below specify the object attribute variables that shall be available in event reports associated with state model transitions:

- Table 14 Variables Required for Substrate Object State Model Transitions
- Table 15 Variables Required for Substrate Location Object State Model Transitions
- Table 16 Variables Required for Batch Location Object State Model Transitions

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13.3 Table 17 provides the definition of Status Variable Data that equipment shall support for the persistent objects defined by this Standard, namely Substrate Location and Batch Location.

13.4 Variables in each table are presented in alphabetical order.

Table 14 Variables Required for Substrate Object State Model Transitions

Variable Name	*Description*	*Access*	*Form*	*Comment*
AcquiredID	Contains the ID read from the substrate. Empty string before the substrate ID is read.	RO	Text.	Empty string if read failed.
AcquiredIDList	A list of AcquiredIDs related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of AcquiredID.	This variable is mandatory only when the equipment provides a substrate reader. When a group of substrates have a related state model transition, only one collection event is triggered. The list length will be equal to the number of substrates involved in the related state model transition.
AdditionalInfo	Specifies additional information for the substrate through name-value pairs.	RO	List of structures consisting of two fields {name, value}	Empty list, if no additional information for the substrate is provided.
AdditionalInfoList	A list of additional information for substrates related by the same Substrate Object State Model state transition that triggered a collection event. Each substrate's additional information is specified through a list of name-value pairs.	RO	Ordered list of AdditionalInfo	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstBatchLocID	Current BatchLocID for the substrate associated with the last event.	RO	Text.	Batch process equipment only. Empty if substrate is not part of a batch.
SubstBatchLocIDList	A list of SubstBatchLocIDs related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstBatchLocID.	Batch process equipment only. When a group of substrates have a related state model transition, only one collection event is triggered.
SubstDestination	Destination Substrate Location for the substrate associated with the last event.	RO	Text.	If empty string, then same as source Substrate Location.

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Variable Name	*Description*	*Access*	*Form*	*Comment*
SubstDestinationList	A list of SubstDestinations related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstDestination.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstHistory	List of history of locations visited for the substrate associated with the last event.	RO	List of structures consisting of Location ID, Time In, and Time Out.	
SubstHistoryList	A list of SubstHistorys related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstHistory.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstID	Identifier of substrate associated with last event.	RO	Text.	
SubstIDList	A list of substrate identifiers related by the same Substrate Object State Model or Substrate Location Object State Model state transition that triggered a collection event with which this variable was associated.	RO	Ordered list of SubstID.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstIDStatus	Status of the substrate ID.	RO	Enumerated: NOT CONFIRMED WAITING FOR HOST CONFIRMED CONFIRMATION FAILED	This variable is mandatory only when the equipment provides a substrate reader.
SubstIDStatusList	A list of SubstIDStatuses related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstIDStatus.	This variable is mandatory only when the equipment provides a substrate reader. When a group of substrates have a related state model transition, only one collection event is triggered. The list length will be equal to the number of substrates involved in the related state model transition.
SubstSubstLocID	Current substrate location for the substrate associated with the last event.	RO	Text.	If batch process equipment, this data is empty while the substrate belongs to a batch.

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Variable Name	*Description*	*Access*	*Form*	*Comment*
SubstLocIDList	A list of substrate location identifiers related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstSubstLocID.	When a group of substrates have a related state model transition, only one collection event is triggered. The list length will be equal to the number of substrates involved in the related state model transition. If batch process equipment, this will be a list of empty strings while the substrate belongs to a batch.
SubstLotID	The LotID of the Substrate Object associated with last event.	RO	Text.	
SubstLotIDList	A list of substrate LotIDs related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstLotID.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstMtrlStatus	Processing quality criteria associated with the last event.	RO	Enumerated (equipment specific).	
SubstMtrlStatusList	A list of MtrlStatus related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstMtrlStatus.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstObjType	Specifies object type of the substrate associated with last event.	RO	Text.	Text value is 'Substrate'
SubstObjTypeList	A list of object types related by the same Substrate Object State Model or Substrate Location Object State Model state transition that triggered a collection event with which this variable was associated.	RO	Ordered list of SubstObjType.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstPosInBatch	Current substrate position in a batch for the substrate associated with the last event.	RO	Text.	Batch process equipment only. Empty if substrate position in a batch is not informed or the substrate is not part of a batch.

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Variable Name	*Description*	*Access*	*Form*	*Comment*
SubstPosInBatchList	A list of SubstPosInBatchs related by same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstPosInBatch.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstProcState	Processing state of the substrate associated with the last event.	RO	Enumerated: NEEDS PROCESSING IN PROCESS PROCESSED ABORTED STOPPED REJECTED	May not be implemented when the equipment does not have capability of process.
SubstProcStateList	A list of substrate process states related by the same Substrate Object State Model state transition that triggered a collection event with which this variable was associated.	RO	Ordered list of SubstProcState.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstSource	Source Substrate Location for the substrate associated with the last event.	RO	Text.	
SubstSourceList	A list of SubstSources related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstSource.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstState	The current state of the substrate associated with the last event.	RO	Enumerated: AT SOURCE AT WORK AT DESTINATION	
SubstStateList	A list of substrate states related by the same Substrate Object State Model state transition that triggered a collection event with which this variable was associated.	RO	Ordered list of SubstState.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstType	Substrate SubstType for the substrate associated with the last event.	RO	Enumerated: wafer, disc, flat panel, reticle	SEMI E5 contains a dictionary of units.
SubstTypeList	A list of SubstTypes related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstType.	When a group of substrates have a related state model transition, only one collection event is triggered.
SubstUsage	Description of the substrate associated with the last event.	RO	Enumerated: PRODUCT TEST FILLER	Indicates how the substrate is used. Additional types defined and added by the equipment.
SubstUsageList	A list of SubstUsages related by the same Substrate Object State Model state transition that triggered a collection event.	RO	Ordered list of SubstUsage.	When a group of substrates have a related state model transition, only one collection event is triggered.

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Table 15 Variables Required for Substrate Location Object State Model Transitions

Variable Name	*Description*	*Access*	*Form*	*Comment*
SubstLocDisableEvents	Indicates if transitions for this substrate location do not generate events to the host.	RO	Boolean	TRUE = Disable FALSE = Enable
SubstLocID	Identifier for substrate location associated with the last event.	RO	Text.	
SubstLocObjType	Object type for substrate location.	RO	Text	Text value is 'SubstLoc'
SubstLocState	State of substrate location associated with the last event.	RO	Enumerated: OCCUPIED UNOCCUPIED	
SubstLocSubstID	Substrate Identifier relevant to the location associated with last event.	RO	Text.	Blank “” when location is UNOCCUPIED.

Table 16 Variables Required for Batch Location Object State Model Transitions

Variable Name	*Description*	*Access*	*Form*	*Comment*
BatchLocDisableEvents	Indicates if transitions for this batch location do not generate events to the host.	RO	Boolean	TRUE = Disable FALSE = Enable
BatchLocID	Identifier for batch location associated with the last event.	RO	Text.	
BatchLocObjType	Object type for batch location.	RO	Text	Text value is 'BatchLoc'
BatchLocState	State of batch location associated with the last event.	RO	Enumerated: OCCUPIED UNOCCUPIED	
BatchSubstIDMap	Ordered list of substrate identifiers corresponding to the positions in a batch (e.g., slot positions of a batch container or a process carrier) associated with the last event.	RO	Ordered list of SubstID.	Ordered list indicates position in the batch and SubstID can be actual substrate identifier, blank “” when location is empty or “filler” when such substrate is used.

Table 17 Status Variables Representing Object Attributes for Persistent Objects

Variable Name	*Description*	*Access*	*Form*	*Comment*
$BatchLocDisableEvents_i$	Indicates if transitions for the ith batch location do not generate events to the host.	RO	Boolean	TRUE = Disable FALSE = Enable
$BatchLocID_i$	Identifier of the ith batch location.	RO	Text.	
$BatchLocState_i$	State of ith batch location.	RO	Enumerated: OCCUPIED UNOCCUPIED	

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Variable Name	*Description*	*Access*	*Form*	*Comment*
$BatchSubstIDMap_i$	Ordered list of substrate identifiers corresponding to the positions in the batch held at the ith batch location.	RO	Ordered list of SubstID.	Ordered list indicates position in the batch and SubstID can be actual substrate identifier, blank “” when position is empty or “filler” when such substrate is used.
$SubstLocDisableEvents_i$	Indicates if transitions for the ith equipment substrate location do not generate events to the host.	RO	Boolean	TRUE = Disable FALSE = Enable
$SubstLocID_i$	Identifier of the ith equipment substrate location.	RO	Text.	
$SubstLocState_i$	State of ith equipment substrate location.	RO	Enumerated: OCCUPIED UNOCCUPIED	
$SubstrLocSubstrID_i$	Substrate ID of the substrate at the ith equipment substrate location, when occupied.	RO	Text.	Blank “” when location is UNOCCUPIED.

14 Additional Events (Equipment with substrate ID reader only)

14.1 *SubstrateIDReaderAvailable Event*

14.1.1 An event shall be generated whenever a substrate ID reader becomes available.

14.2 *SubstrateIDReaderUnavailable Event*

14.2.1 An event shall be generated whenever a substrate ID reader becomes unavailable.

15 Well-Known Names

15.1 *Background*

15.1.1 Implementers of this Specification may choose any names for variables, collection events and alarms. For example, SEMI E90 defines a collection event for Substrate Location State Model Transition 1. Implementers might define the CENAME for this collection event as E90_Loc_Unoccupied2Occupied, SLTrans1, SubstrateLocationUnoccupiedToOccupied, SLSM01 or CollectionEvent17. All of these CENAME are valid.

15.1.2 For end users, it can be useful to map the names for variables, collection events and alarms in a specific implementation to the variables, collection events and alarms defined in this Specification. Well-known names are standardized names for the variables, collection events and alarms defined in this Specification.

15.1.3 Equipment implementing this Specification can publish a map between the names in their implementation and the well-known names defined in this section using a SEMI E172 SECS Equipment Data Dictionary (SEDD) file. The map can also be published in the SEMI E30 documentation.

15.1.4 New implementations can use well-known names as the SECS item name (CENAME, SVNAME, etc.).

15.1.5 The list of well-known variables, collection events and alarms in this section does not intend to imply that implementations of this specification must include all the listed variables, collection events and alarms.

15.1.6 For completeness, well-known tables for variables, collection events and alarms included yet deliberately left empty when no applicable variable, collection event or alarm is defined in this Specification.

15.1.7 Well-known names are case-sensitive.

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15.1.8 A “<#>” in the well-known name means that there can be multiple instances. Sufficient leading zeros should be added to allow logical alphabetical sorting. For example, E90:SubstLocID.0001, E90:SubstLocID.0002, …, E90:SubstLocID.1000 if there are up to 9999 instances.

15.2 *Well-Known Names for Variables*

15.2.1 This section defines well-known names for variables currently defined in this Specification.

15.2.2 Refer to SEMI E5 for more information on the variable classes.

15.2.3 When a variable corresponds to an object’s attribute, the well-known name uses the naming convention ‘E90:<object name>:<attribute name>’ where ‘<object name>’ is the name of the object without spaces, ‘<attribute name>’ is name of the corresponding attribute without spaces.

Table 18 Well-Known Names for Variable

Variable	*Variable Class*	*Well-Known Name*
AcquiredID	DV	E90:Substrate:AcquiredID
SubstBatchLocID	DV	E90:Substrate:BatchLocID
SubstObjType	DV	E90:Substrate:SubstObjType
SubstLotID	DV	E90:Substrate:LotID
SubstMtrlStatus	DV	E90:Substrate:MaterialStatus
SubstID	DV	E90:Substrate:ObjID
SubstDestination	DV	E90:Substrate:SubstDestination
SubstHistory	DV	E90:Substrate:SubstHistory
SubstIDStatus	DV	E90:Substrate:SubstIDStatus
SubstSubstLocID	DV	E90:Substrate:SubstLocID
SubstPosInBatch	DV	E90:Substrate:SubstPosInBatch
SubstProcState	DV	E90:Substrate:SubstProcState
SubstSource	DV	E90:Substrate:SubstSource
SubstState	DV	E90:Substrate:SubstState
SubstType	DV	E90:Substrate:SubstType
SubstUsage	DV	E90:Substrate:SubstUsage
AcquiredIDList	DV	E90:AcquiredIDList
SubstBatchLocIDList	DV	E90:SubstBatchLocIDList
SubstLotIDList	DV	E90:SubstLotIDList
SubstMtrlStatusList	DV	E90:SubstMtrlStatusList
SubstIDList	DV	E90:SubstIDList
SubstDestinationList	DV	E90:SubstDestinationList
SubstHistoryList	DV	E90:SubstHistoryList
SubstIDStatusList	DV	E90:SubstIDStatusList
SubstLocIDList	DV	E90:SubstLocIDList
SubstObjTypeList	DV	E90:SubstObjTypeList
SubstPosInBatchList	DV	E90:SubstPosInBatchList
SubstProcStateList	DV	E90:SubstProcStateList
SubstSourceList	DV	E90:SubstSourceList
SubstStateList	DV	E90:SubstStateList
SubstTypeList	DV	E90:SubstTypeList
SubstUsageList	DV	E90:SubstUsageList
SubstLocDisableEvents	DV	E90:SubstrateLocation:SubstLocDisableEvents

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Variable	*Variable Class*	*Well-Known Name*
SubstLocState	DV	E90:SubstrateLocation:SubstLocState
SubstLocSubstID	DV	E90:SubstrateLocation:SubstLocSubstID
SubstLocID	DV	E90:SubstrateLocation:ObjID
SubstLocObjType	DV	E90:SubstrateLocation:SubstLocObjType
BatchLocDisableEvents	DV	E90:BatchLocation:BatchLocDisableEvents
BatchLocState	DV	E90:BatchLocation:BatchLocState
BatchSubstIDMap	DV	E90:BatchLocation:BatchSubstIDMap
BatchLocID	DV	E90:BatchLocation:ObjID
BatchLocObjType	DV	E90:BatchLocation:BatchLocObjType
$BatchLocDisableEvents_i$	SV	E90:BatchLocation:BatchLocDisableEvents.<#>
$BatchLocID_i$	SV	E90:BatchLocID.<#>
$BatchLocObjType_i$	SV	E90:BatchLocation:BatchLocObjType.<#>
$BatchLocState_i$	SV	E90:BatchLocState.<#>
$BatchSubstIDMap_i$	SV	E90:BatchSubstIDMap.<#>
$SubstLocDisableEvents_i$	SV	E90:SubstrateLocation:SubstLocDisableEvents.<#>
$SubstLocID_i$	SV	E90:SubstrateLocation:SubstLocID.<#>
$SubstLocObjType_i$	SV	E90:SubstrateLocation:SubstLocObjType.<#>
$SubstLocState_i$	SV	E90:SubstrateLocation:SubstLocState.<#>
$SubstrLocSubstrID_i$	SV	E90:SubstrateLocation:SubstLocSubstID.<#>
SubstrateReaderEnabled	EC	E90:SubstrateReaderEnabled

NOTE 3: The well-known name for Variable SubstrLocSubstrID is intentionally spelt differently from the variable name to maintain consistency with short form of substrate used in the other well-known names.

15.3 *Well-Known Names for Collection Events*

15.3.1 This section describes well-known names for collection events currently defined in the Standard.

15.3.2 When collection events correspond to a state model, the well-known name uses the naming convention ‘E90:<state model name>:<transition number>:<from state>-<to state>’ where ‘<state model name>’ is the name of the state model without spaces, ‘<transition number>’ is the respective state model transition number using 3 digits and leading zeros when necessary, ‘<from state>’ is the name of the initial state without spaces, and ‘<to state>’ is the name of the ending state without spaces. The name ‘NoState’ is used to describe the initial state when created and also the final transition when destroyed.

Table 19 Collection Event Well-Known Names

Collection Event	*Well-Known Name*
Substrate Object State Model Transition 1	E90:Substrate:001:NoState-AtSource
Substrate Object State Model Transition 2	E90:Substrate:002:AtSource-AtWork
Substrate Object State Model Transition 3	E90:Substrate:003:AtWork-AtSource
Substrate Object State Model Transition 4	E90:Substrate:004:AtWork-AtWork
Substrate Object State Model Transition 5	E90:Substrate:005:AtWork-AtDestination
Substrate Object State Model Transition 6	E90:Substrate:006:AtDestination-AtWork
Substrate Object State Model Transition 7	E90:Substrate:007:NoState
Substrate Object State Model Transition 8	E90:Substrate:008:AtDestination-AtSource
Substrate Object State Model Transition 9	E90:Substrate:009:NoState
Substrate Object State Model Transition 10	E90:Substrate:010:NoState-NeedsProcessing

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Collection Event	*Well-Known Name*
Substrate Object State Model Transition 11	E90:Substrate:011:NeedsProcessing-InProcess
Substrate Object State Model Transition 12	When implemented as six collection events: E90:Substrate:012:InProcess-Processed E90:Substrate:012:InProcess-Rejected E90:Substrate:012:InProcess-Stopped E90:Substrate:012:InProcess-Aborted E90:Substrate:012:InProcess-Lost E90:Substrate:012:InProcess-Skipped When implemented as one collection event: E90:Substrate:012:ProcessingComplete
Substrate Object State Model Transition 13	E90:Substrate:013:InProcess-NeedsProcessing
Substrate Object State Model Transition 14	When implemented as two collection events: E90:Substrate:014:NeedsProcessing-Lost E90:Substrate:014:NeedsProcessing-Skipped When implemented as one collection event: E90:Substrate:012:ProcessingComplete
Substrate Object State Model Transition 16	E90:Substrate:016:NoState-NotConfirmed
Substrate Object State Model Transition 17	E90:Substrate:017:NotConfirmed-Confirmed
Substrate Object State Model Transition 18	E90:Substrate:018:NotConfirmed-WaitingForHost
Substrate Object State Model Transition 19	E90:Substrate:019:NotConfirmed-WaitingForHost
Substrate Object State Model Transition 20	E90:Substrate:020:WaitingForHost-Confirmed
Substrate Object State Model Transition 21	E90:Substrate:021:WaitingForHost-ConfirmationFailed
Substrate Location State Model Entry Transition	When implemented as two collection events: E90:SubstrateLocation:000:NoState-Unoccupied E90:SubstrateLocation:000:NoState-Occupied When implemented as one collection event: E90:SubstrateLocation:000:NoState
Substrate Location State Model Transition 1	E90:SubstrateLocation:001:Unoccupied-Occupied
Substrate Location State Model Transition 2	E90:SubstrateLocation:002:Occupied-Unoccupied
Batch Location State Model Entry Transition	E90:BatchLocation:000:NoState-Occupied E90:BatchLocation:000:NoState-Unoccupied
Batch Location State Model Transition 1	E90:BatchLocation:001:Unoccupied-Occupied
Batch Location State Model Transition 2	E90:BatchLocation:002:Occupied-Unoccupied
SubstrateIDReaderAvailable	E90:SubstrateIDReaderAvailable
SubstrateIDReaderUnavailable	E90:SubstrateIDReaderUnavailable

15.4 *Well-Known Names for Alarms*

15.4.1 This section describes well-known names for alarms currently defined in the Standard.

15.4.2 This table is deliberately left empty as there are no explicit alarms defined in this Standard.

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Table 20 Well-Known Names for Active Alarms

Alarm	*Well-Known Name*

16 Requirements for Compliance

16.1 Table 21 provides a checklist for Substrate Tracking (STS) compliance.

Table 21 STS Compliance Statement

Fundamental STS Requirements	*STS Section*	*Implemented*	*STS Compliant*
Substrate Tracking	8 (except 8.4)	☐ Yes ☐ No	☐ Yes ☐ No
Substrate Object and State Model	9	☐ Yes ☐ No	☐ Yes ☐ No
Substrate Location Object and State Model	10	☐ Yes ☐ No	☐ Yes ☐ No
Service Message Implementation (except host-initiated Register (Create) Substrate and Remove Substrate)	12	☐ Yes ☐ No	☐ Yes ☐ No
Variable Data	13	☐ Yes ☐ No	☐ Yes ☐ No
Events	8.4	☐ Yes ☐ No	☐ Yes ☐ No
Additional STS Capabilities	*STS Section*	*Implemented*	*STS Compliant*
Host-Initiated Register (Create Substrate)	12.1.1	☐ Yes ☐ No	☐ Yes ☐ No
Host-Initiated Remove Substrate		☐ Yes ☐ No	☐ Yes ☐ No
Batch Location Object and State Model	11	☐ Yes ☐ No	☐ Yes ☐ No
UpdateSubstrateObject service	12	☐ Yes ☐ No	☐ Yes ☐ No
Substrate Reading Status	9.3	☐ Yes ☐ No	☐ Yes ☐ No
‘SubstIDStatus’ substrate object attribute	9.5	☐ Yes ☐ No	☐ Yes ☐ No
‘AdditionalInfo’ substrate object attribute	9.5	☐ Yes ☐ No	☐ Yes ☐ No
ProceedWithSubstrate service	12.2	☐ Yes ☐ No	☐ Yes ☐ No
CancelSubstrate service	12.2	☐ Yes ☐ No	☐ Yes ☐ No
Additional Events	14	☐ Yes ☐ No	☐ Yes ☐ No

17 Related Documents

17.1 *Subordinate Standards*

SEMI E90.1 — Specification for SECS-II Protocol Substrate Tracking

17.2 *SEMI Standards and Safety Guidelines*

SEMI E87 — Specification for Carrier Management (CMS)

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APPENDIX 1
WELL-KNOWN NAMES FOR RETIRED VARIABLES, COLLECTION EVENTS and ALARMS

NOTICE: The material in this Appendix is an official part of SEMI E90 and approved by full Letter Ballot procedures on December 14, 2024.

A1-1 Background

A1-1.1 This Appendix describes well-known names for items previously defined in this Standard that are not defined in the current revision. They are provided for backwards compatibility.

A1-1.2 For completeness, well-known tables for variables, collection events and alarms included yet deliberately left empty when no applicable variable, collection event or alarm is defined in this Specification.

A1-1.3 The Last Revision column captures the last revision of this Standard where the item was defined.

A1-2 Well-Known Names for Retired Variables

A1-2.1 This section describes well-known names for variables previously defined in the Standard.

A1-2.2 Refer to SEMI E5 for more information on the variable classes.

Table A1-1 Well-Known Names for Retired Variables

Variable	*Variable Class*	*Well-Known Name*	*Last Revision*

A1-3 Well-Known Names for Retired Collection Events

A1-3.1 This section describes well-known names for collection events previously defined in the Standard.

Table A1-2 Well-Known Names for Retired Collection Events

Collection Event	*Well-Known Name*	*Last Revision*

A1-4 Well-Known Names for Retired Alarms

A1-4.1 This section describes well-known names for alarms previously defined in the Standard.

Table A1-3 Well-Known Names for Retired Alarms

Alarm	*Well-Known Name*	*Last Revision*

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RELATED INFORMATION 1

NOTICE: This Related Information is not an official part of SEMI E90 and was derived from the work of the Information & Control Global Technical Committee. This Related Information was approved for publication by full letter ballot procedures on January 14, 2000.

R1-1 Substrate State Model Implementation

R1-1.1 The Substrate Object State Model defined in ¶ 9.1 is a generic representation of the substrate object state model that includes standard transitions for various equipment types.

R1-1.2 Some equipment may not be able to implement all of the transitions defined in this model. For example, the transition from AT WORK to AT SOURCE (Transition Number 3) represents the case where the substrate returns to the original (source) location, typically for further processing. If the equipment is unable to implement this return, then the transition would never occur, and it should be omitted from the implemented state model.

R1-1.3 The state model shall be implemented based on the definition of the state model; however, substates and transitions that cannot occur for a particular equipment design may be omitted where the model allows.

R1-2 Example of Substrate Object State Model for Simple Equipment

R1-2.1 This section shows an example of Substrate Object State Model for simple equipment (Figure R1-1).

R1-2.2 The equipment transports the substrate only in a forward direction, so that the Substrate Transport Substate takes straight forward transitions from AT SOURCE to AT WORK and then to AT DESTINATION. For example, the wirebonder removes leadframes from a magazine and carries them on a transport mechanism but is unable to return them to the magazine.

R1-2.3 Also, the equipment processes the substrate only in forward steps so that the Substrate Processing Substate takes straight forward transitions from NEED PROCESSING to IN PROCESS and then to PROCESSING COMPLETE. The REJECTED state is not applied to this equipment.

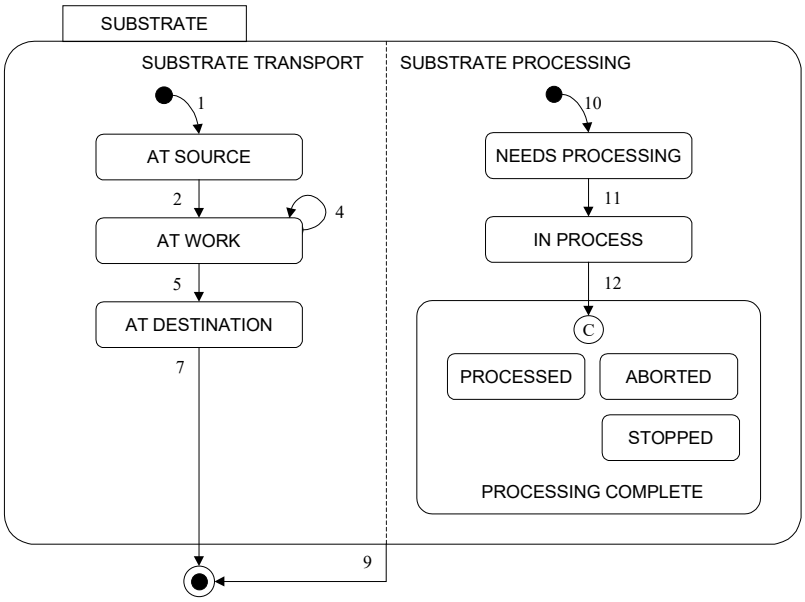

Figure R1-1
Substrate State Model for Simple Equipment

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